

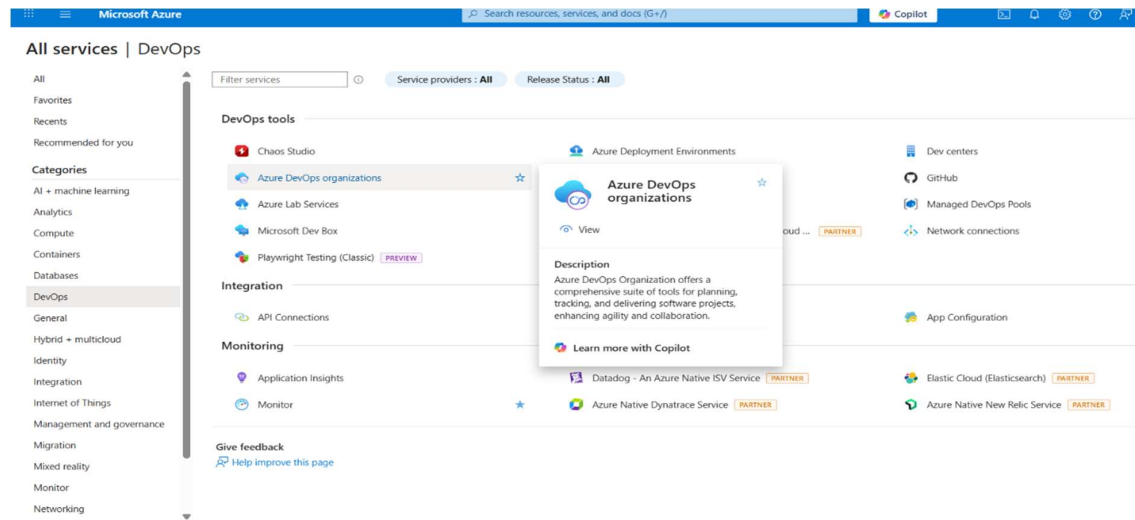
Azure DevOps — CI/CD Pipeline Creation (Step-by-Step)

Purpose:

This document shows how to create a simple CI/CD pipeline in Azure DevOps using a Git repository and a self-hosted agent. Each step has wording and a placeholder where you should insert your screenshot.

Step 1 — Open Azure portal and go to Azure DevOps

Sign in to the Azure portal (<https://portal.azure.com>). From the left-hand menu search for and open **Azure DevOps** to manage organizations and projects.



Step 2 — Click My Azure DevOps Organizations

In the Azure DevOps landing page click **My Azure DevOps organizations** to view existing DevOps organizations associated with your account.

Azure DevOps

Plan smarter, collaborate better, and ship faster with a set of modern dev services

[My Azure DevOps Organizations](#)

[Get started using Azure DevOps](#)

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Step 3 — Create a new organization

Click **Create new organization** to start a new Azure DevOps organization.

Microsoft

Sivagami G Sign out



Sivagami G
sivagamisiva774@gmail.com
[Edit profile](#)

Microsoft account

India
sivagamisiva774@gmail.com

Visual Studio Dev Essentials
Get everything you need to build and deploy your app on any platform.
[Use your benefits](#)



Get started with Azure DevOps
Plan better, code together, ship faster with Azure DevOps
[Create new organization](#)

Step 4— Fill organization details and continue

Fill in the required organization details (name, region) and click **Continue** to proceed.

 Azure DevOps

sivagamisiva774@gmail.com [Switch directory](#)

Almost done...

Name your Azure DevOps organization *

dev.azure.com/ sivagamisiva774

We'll host your projects in

India

Enter the characters you see

New Audio



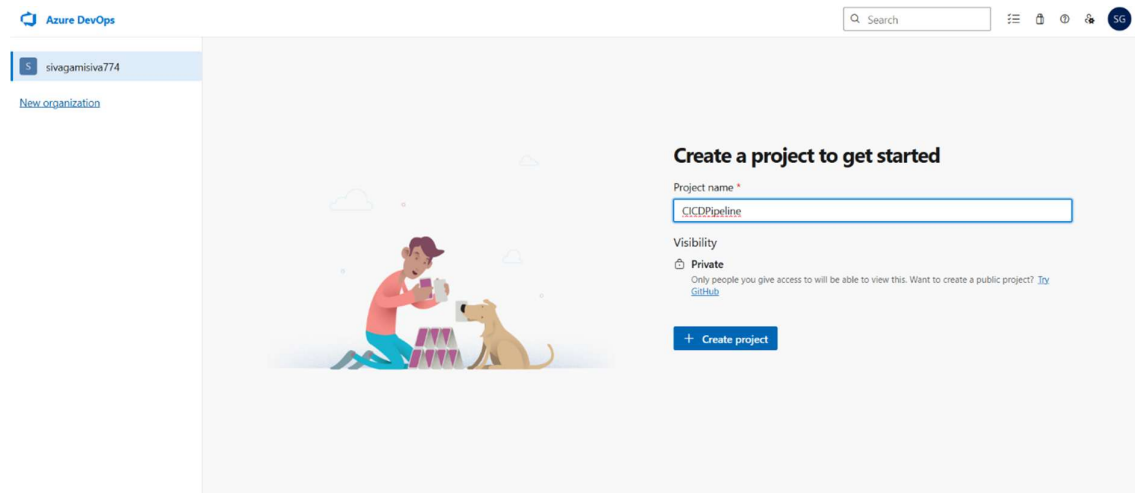
NSY4KW

abc

[Continue](#)

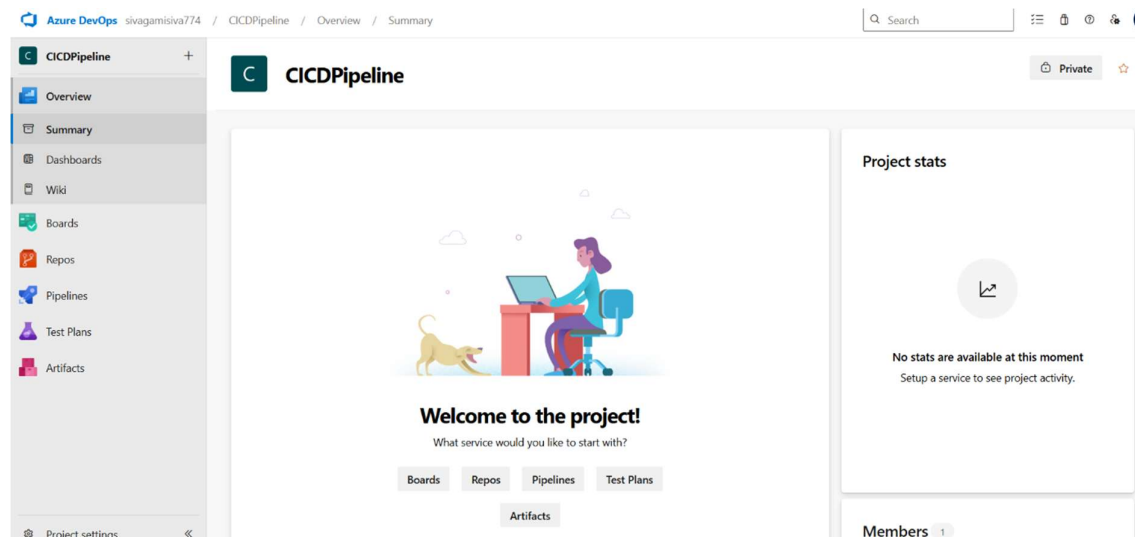
Step 5 — Create a project

After the organization is created you will be directed to the **Create project** page. Provide a project name (for example CI-CD-Pipeline) and click **Create project**.



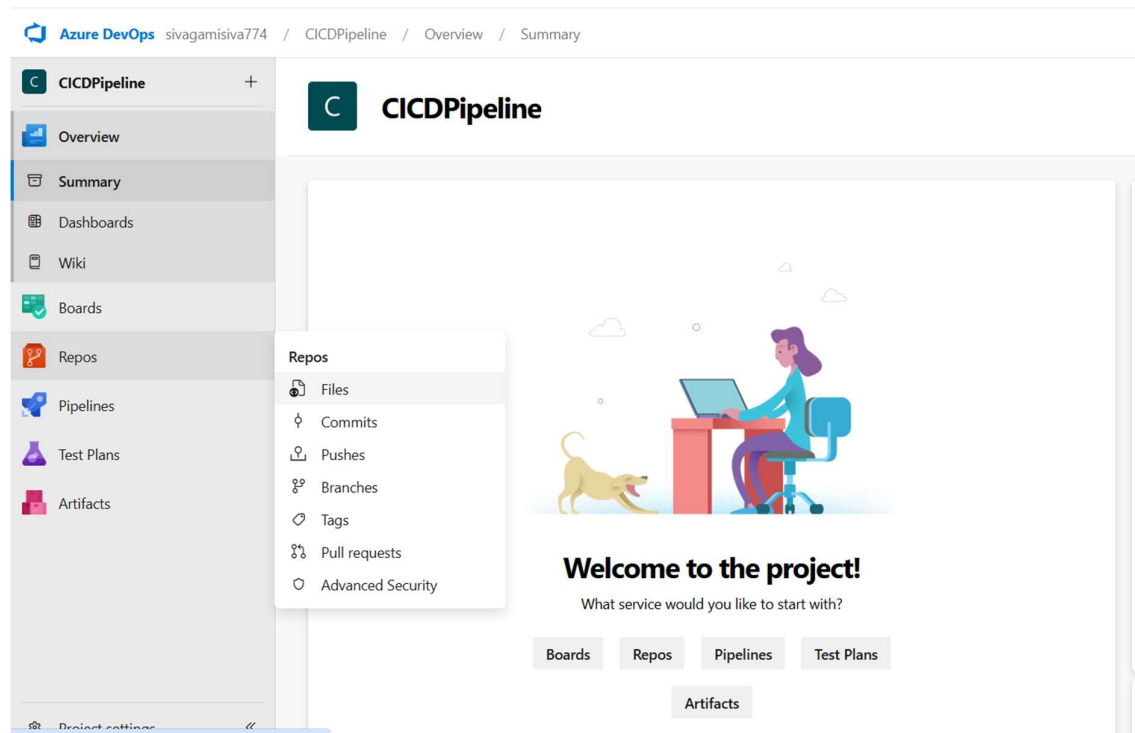
Step 6 — Azure DevOps home page (project view)

Once the project is created you will be redirected to the project home page where you can access Repos, Pipelines, Boards, and other features.



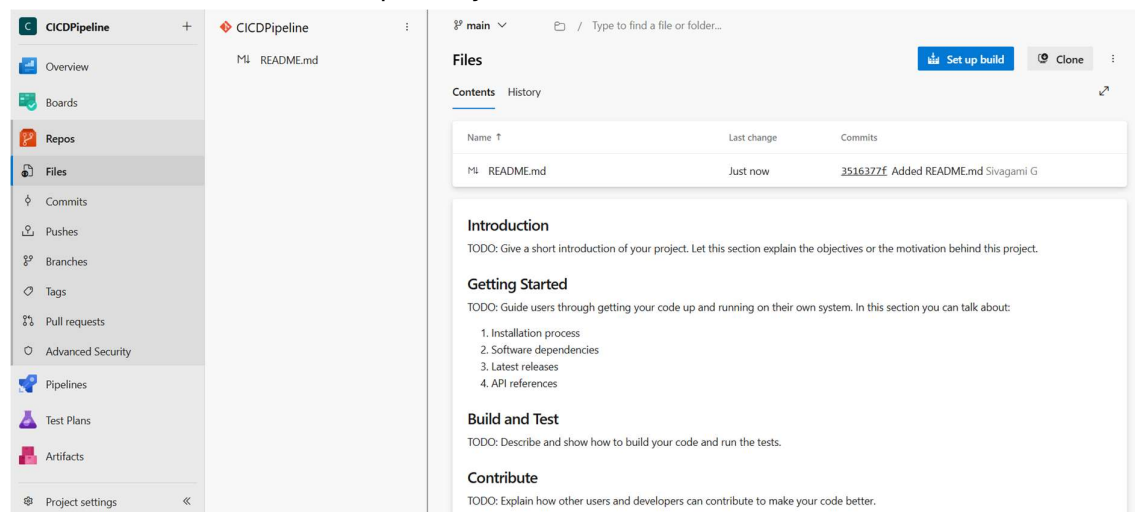
Step 7 — Go to Repos → Files

From the left-hand menu select **Repos** then **Files** to view or add files to the repository.



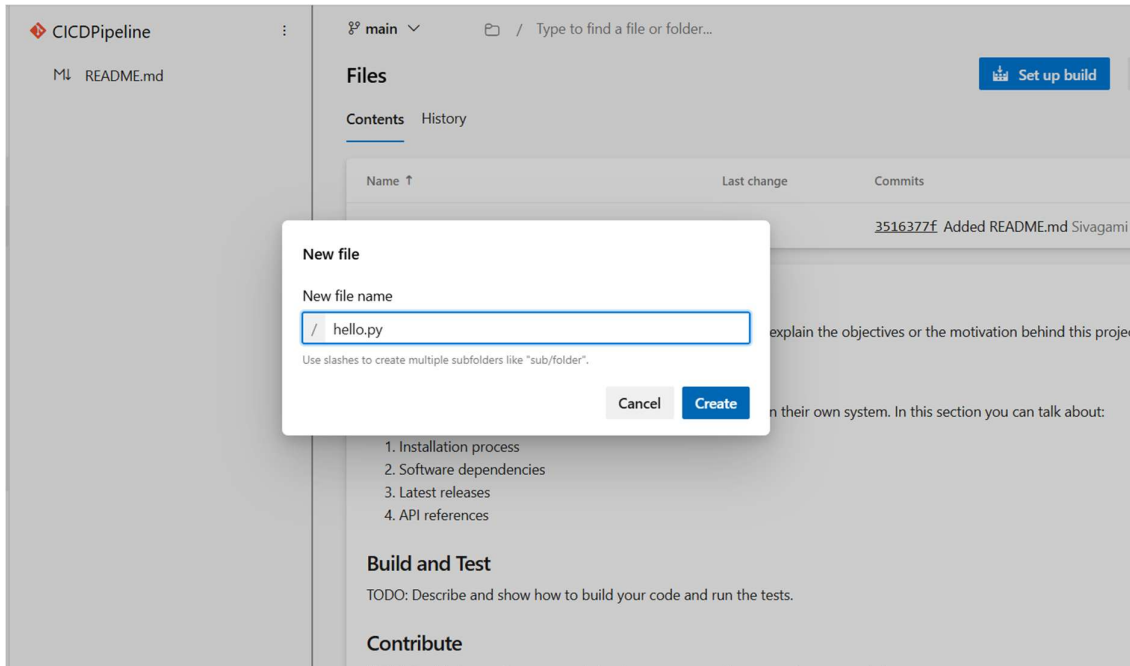
Step 8 — README file page

The Files view shows the repository contents including the README.md. You can edit this or add new files to the repository.



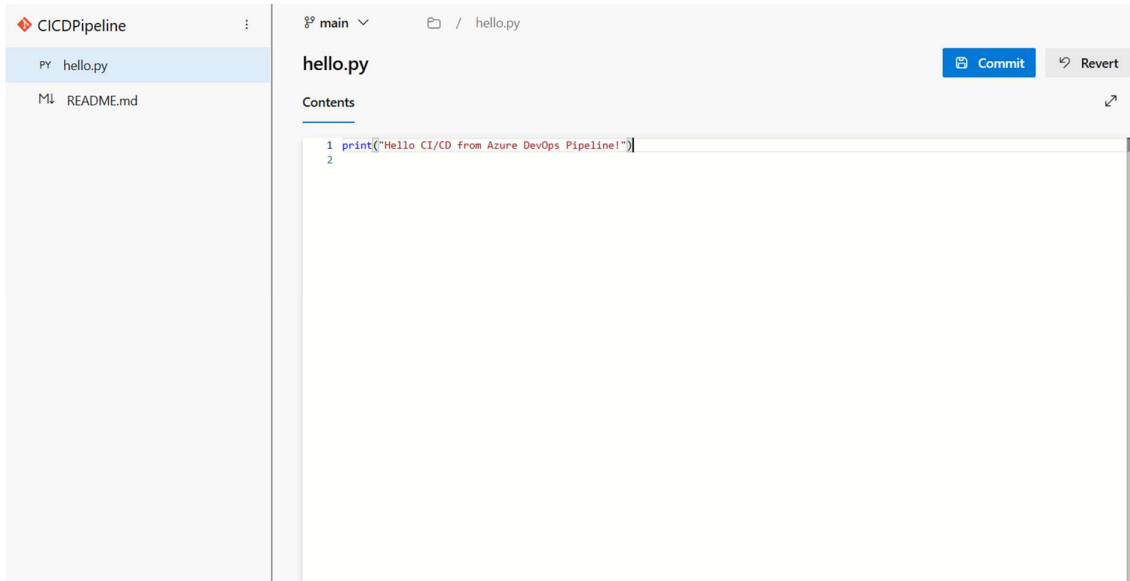
Step 9 — Create a new file hello.py

In your project repository click the three dots (ellipsis) next to the repo name and select **Create new file**. Name the file `hello.py`.



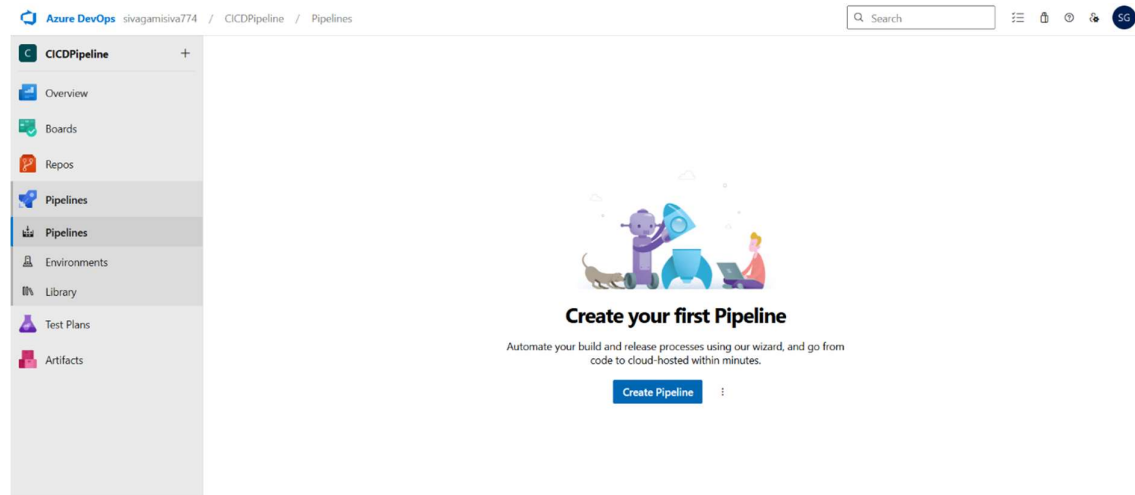
Step 10 — Add contents to hello.py and commit

Paste your Python code into `hello.py`. Add a meaningful commit message in the commit pane on the right and click **Commit** to save the file to the repository.



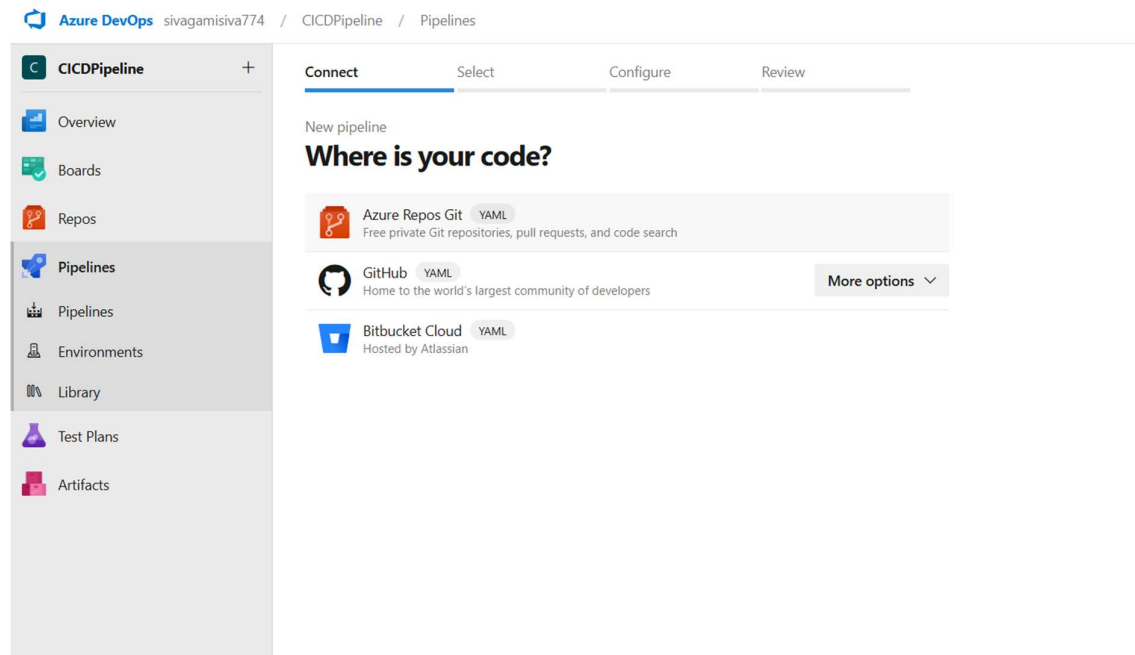
Step 11 — Go to Pipelines → Create Pipeline

From the left menu select **Pipelines** then click **Create pipeline** to start the pipeline creation wizard.



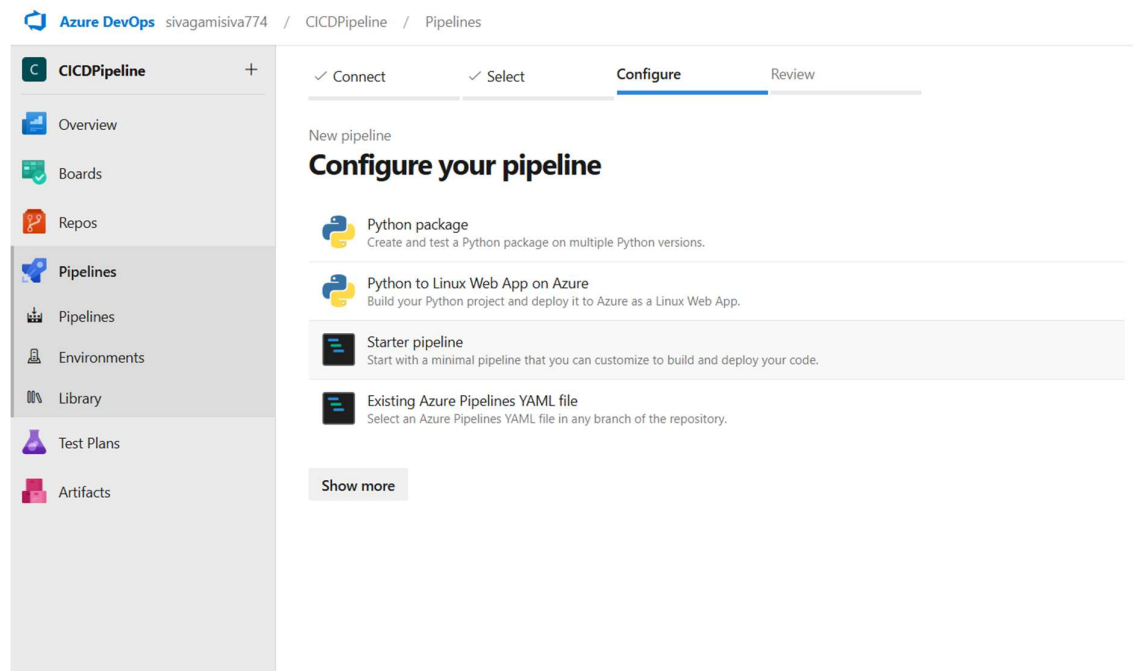
Step 12 — Select GitHub as the source

Select **GitHub** (or the Git provider where your repo is hosted) as the code source. Authenticate if prompted.



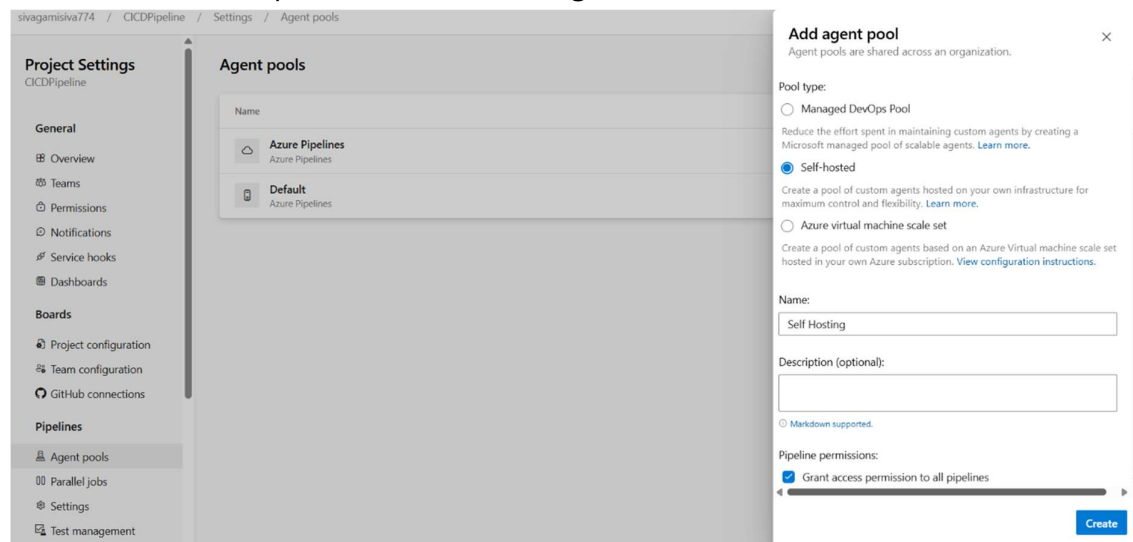
Step 13 — Start with a starter pipeline and review

Choose **Starter pipeline** to generate a basic azure-pipelines.yml. Review the generated YAML and modify it if necessary. Save or run the pipeline to create it.



Step 14 — Configure Agent Pools: create a self-hosted pool

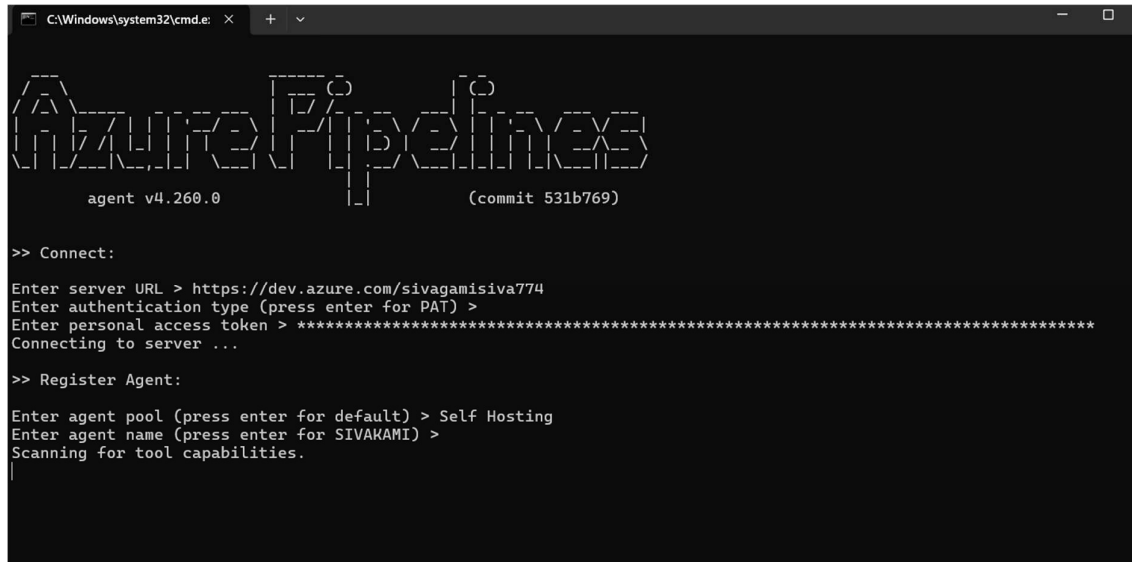
Navigate to **Project settings** → **Agent pools**. Choose **New agent pool** (or **Add pool**) and create a self-hosted pool named SelfHosting.



Step 15 — Configure the agent on your machine (PowerShell)

On the machine that will host the agent, download the agent package from the agent pool page. Extract and open PowerShell (or terminal) in the agent folder and run the configuration script. Example (Windows):

The config.cmd step registers the agent with Azure DevOps and run.cmd starts it.



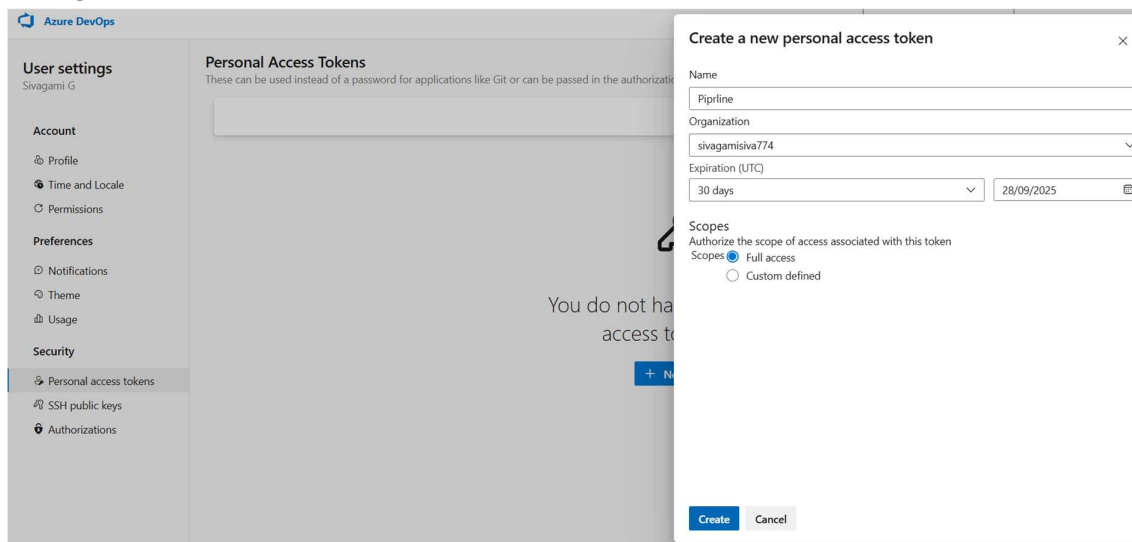
```
C:\Windows\system32\cmd.exe: x + v
agent v4.260.0 (commit 531b769)

>> Connect:
Enter server URL > https://dev.azure.com/sivagamisiva774
Enter authentication type (press enter for PAT) >
Enter personal access token > *****
Connecting to server ...

>> Register Agent:
Enter agent pool (press enter for default) > Self Hosting
Enter agent name (press enter for SIVAKAMI) >
Scanning for tool capabilities.
```

Step 16 — Create a Personal Access Token (PAT)

From your user profile (top-right avatar) open **User settings** → **Personal access tokens** and create a new token. Use a descriptive name, set an expiration, and give the token the necessary scopes. Copy the token securely — you will need it during agent configuration.



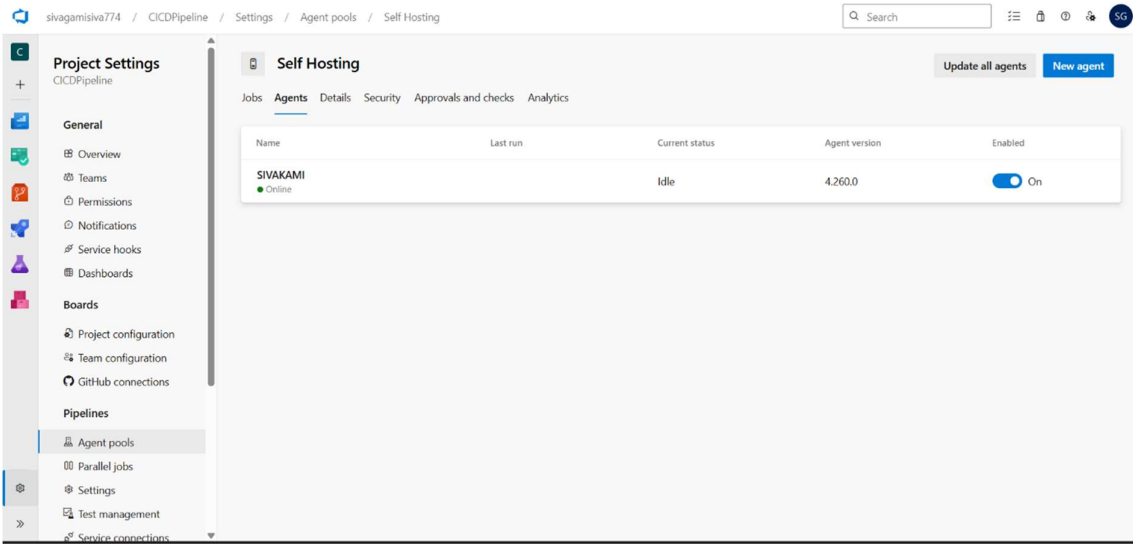
Step 17 — Run the agent so it connects to Azure DevOps

After configuring the agent with the PAT, run the agent script to start the agent. On Windows use `.\run.cmd`. The agent will contact Azure DevOps and appear in the agent pool as online.

```
C:\Windows\system32\cmd.e: X + v
Scanning for tool capabilities.
Connecting to the server.
2025-08-29 06:35:41Z: Listening for Jobs
2025-08-29 06:37:00Z: Running job: Job
2025-08-29 06:37:24Z: Job Job completed with result: Succeeded
2025-08-29 06:39:41Z: Running job: Job
2025-08-29 06:40:01Z: Job Job completed with result: Succeeded
```

Step 18 — Confirm the agent shows online in Azure DevOps

Return to **Project settings** → **Agent pools** and verify that the newly registered agent is listed and its status is **Online**.



Step 19 — Run the pipeline and confirm success

Trigger the pipeline by clicking **Run** (or commit to the branch that triggers the pipeline). Monitor the pipeline execution and confirm that the build and any tasks complete successfully.

CICDPipeline

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

#20250829.6 • Update azure-pipelines.yml for Azure Pipelines

Success

Run new

This run is being retained as one of 3 recent runs by main (Branch).

View retention leases

Summary

Code Coverage

Manually run by Sivagami G

Repository and version

Time started and elapsed

Related

Tests and coverage

CICDPipeline

Just now

0 work items

Get started

main

22s

0 artifacts

Azure DevOps

sivagamisiva774

CICDPipeline

Pipelines

CICDPipeline

Search

SG

CICDPipeline

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

CICDPipeline

Edit

Run pipeline

Runs

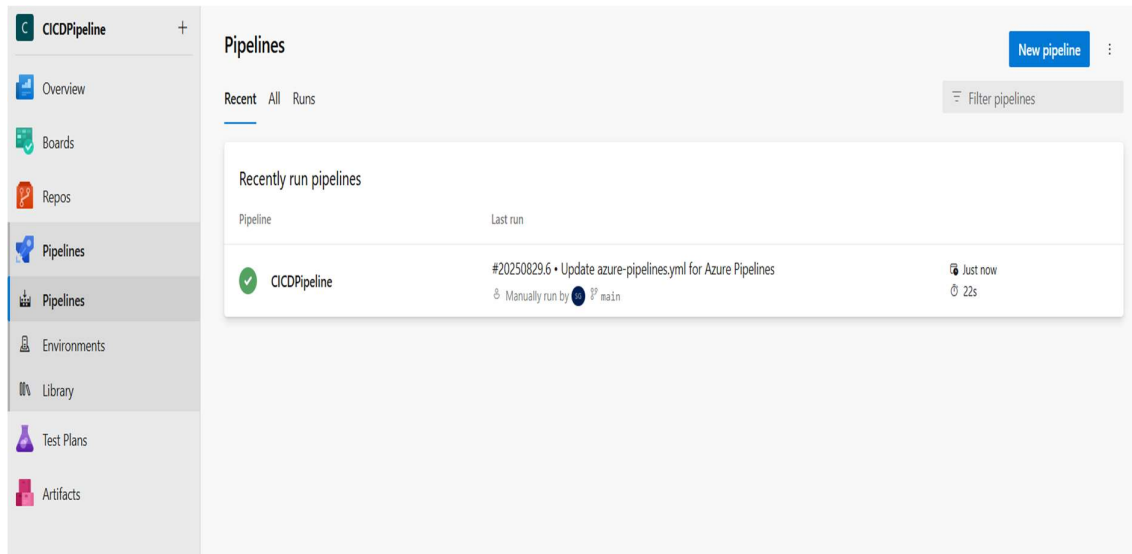
Branches

Analytics

Description	Stages	
#20250829.6 • Update azure-pipelines.yml for Azure Pipelines Manually run by main b64fa868	✓	Just now 22s
#20250829.5 • Update azure-pipelines.yml for Azure Pipelines Manually run by main b64fa868	✓	4m ago 26s
#20250829.4 • Update azure-pipelines.yml for Azure Pipelines Manually run by main b64fa868	✗	16m ago <1s
#20250829.3 • Update azure-pipelines.yml for Azure Pipelines Individual CI for main b64fa868	✗	16m ago <1s
#20250829.2 • Set up CI with Azure Pipelines Manually run by main 43da7d46	✗	22m ago <1s
#20250829.1 • Set up CI with Azure Pipelines Individual CI for main 43da7d46	✗	25m ago <1s

Step 20 — View recently run pipelines

From the **Pipelines** menu you can see recent runs, logs, and results. The right side of the Pipelines page lists recent pipelines and their statuses.



Sample hello.py

```
print("Hello from Azure DevOps pipeline!")
```

azure-pipelines.yml

trigger:

- main

pool:

name: 'SelfHosting'

steps:

- checkout: self

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x'

- script: python hello.py

displayName: 'Run hello.py'

Conclusion:

This documentation demonstrated step-by-step creation of a CI/CD pipeline in Azure DevOps, from organization setup to running pipelines. With proper configuration of repos, pipelines, and agents, automated builds and deployments ensure efficiency, consistency, and reliability in software delivery.
